A Modal Description of Dynamic Wake Meandering

Nicholas Hamilton,[a] Paula Doubrawa, Patrick Moriarty, Stefano Letizia, and Regis Thedin

National Renewable Energy Laboratory (NREL), Golden, Colorado, USA

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Horizontal scans from nacelle-mounted lidar systems provide time series of wake measurements during ~~three~~ different atmospheric inflow conditions, from which we describe the coherent turbulent structures that contribute to wake meandering through proper orthogonal decomposition. Subsets of modes are used to make low-order flow field reconstructions in a combinatorial sense, yielding more than 16,000 estimates of meandering for each of the inflow cases. A regression test using the range of reconstructed flow statistics identifies the modes that contribute most to the accurate description of wake meandering. Spectra defined from each mode coefficient time series highlight the dominant Strouhal number associated with each coherent turbulent structure, and suggest that the lowest ranking modes do not necessarily contribute most to the accurate representation of wake meandering. Instead, some modes appear to have no influence on meandering dynamics, and still others consistently work against the accurate description of wake meandering in reduced-order models. No consistent relationship is revealed between characteristic frequencies for each mode and for either the inflow or the lidar measurements, suggesting that a more complex relationship between wake and inflow turbulence may be needed to accurately describe wake meandering.

[a]Electronic mail: nicholas.hamilton@nrel.gov

I. INTRODUCTION

Wake meandering, characterized by large-scale, quasiperiodic, low-frequency oscillations of the entire wind turbine wake, plays a crucial role in wind turbine loads, controller set point uncertainty, and power quality. Meandering significantly impacts power production and fatigue loading of downstream turbines in wind farms.[1–3] Despite its importance, accurately representing wake meandering in numerical models remains challenging at any level of fidelity, making any additional characterization of the phenomenon valuable for improving wind farm performance and reliability.

Two prevailing hypotheses attempt to explain the origin of wake meandering. The externally driven hypothesis considers the wake's momentum deficit as a passive tracer advected by large-scale atmospheric turbulence.[1] This theory suggests that only turbulent structures larger than the rotor diameter significantly contribute to wake meandering.[1,4–8]

Alternatively, the internally driven hypothesis proposes that dynamic interactions between turbulent structures within the wake itself, such as tip vortices and hub/root vortices, generate pressure fluctuations substantial enough to displace the wake.[9–14] Also within the internally-driven set of hypotheses is that wake meandering may result from a fundamental shear instability that can be excited by multiple mechanisms.[15,16]

Our ability to measure wind turbine wakes in the field has improved tremendously over the past decade. Ten years ago, some of the first dynamic wake measurements were collected by modifying a prototype lidar originally designed for vertical conical scanning.[17,18] These early measurements, despite their limitations, were used to validate basic assumptions of the dynamic wake meandering (DWM) model.[1,7] Today, lidar technology allows us to observe wake dynamics for wind turbines of any size, with the capability to scan in any direction spanwise and vertically, reach downstream distances on the order of kilometers, and capture wake meandering, the downstream trajectory of wake centers, expansion, and large-scale turbulent structures.[2] Nacelle-mounted lidars have emerged as a powerful tool for real-time characterization of inflow and wakes of utility-scale wind turbines.[19] Doppler lidars offer unique capabilities for wind energy research, including real-time measurements for turbine and farm control, long-term performance assessment, high-resolution data assimilation for numerical simulations, and detailed wake characterization.[18,20–26] provide better characterization of the wind resource at relevant heights and allow effortless steering onto the wind direction for yaw-controlled turbines.

Several recent studies leverage modern lidar technology to characterize turbulence in the wake. Depending on the instrument deployed, lidars typically measure in vertical planes at fixed downstream distances,[25,27,28] or in horizontal planes perpendicular to the rotor at hub height.[29,30] These recent, lidar-based studies have primarily looked at the temporal evolution of turbulence by identifying the wake within each scan and tracking its centerline position over time. The centerline is then used to quantify wake meandering for the sake of wake characterization[27] or model validation.[25]

To investigate under what conditions we can expect the internally-driven hypothesis to be the dominant mechanism driving wake meandering, we employ Proper Orthogonal Decomposition (POD). This method is well-suited for identifying coherent turbulence structures in flow fields[31] and has been applied to wind fields for over fifteen years.[32–36] POD seeks the variance-maximizing structures that describe the turbulent kinetic energy in the flow, making it an ideal tool for quantifying potential meandering-inducing structures in the wake. Our approach involves developing a reduced-order model (ROM) using the POD-derived structures. If these structures are shown to contribute significantly to wake meandering, the ROM should reproduce the meandering behavior seen in the measurement data to a high level of fidelity. Conversely, a ROM developed without these structures should fail to accurately reproduce wake meandering.

Through the use of POD, we ~~will~~ estimate the meandering length scales and frequencies in our dataset and relate them to DWM modeling assumptions. The body of past experimental and numerical work typically characterizes the meandering frequency (f_m) in the context of a Strouhal number, $St = f_m D/U_{\text{hub}}$ defined in terms of the wind turbine rotor diameter (D) and hub-height inflow velocity (U_{hub}). Meandering frequencies are typically reported in the range of $0.1 \leq St \leq 0.35$, both in laboratory and field observations.[2,37,38] If the internally-driven hypothesis for wake meandering is correct, the modes that contribute most to meandering should have characteristic frequencies higher than the inflow Strouhal number. Conversely, cases where the externally driven hypothesis is correct should favor structures with frequencies below the inflow Strouhal number.

DWM models[1,8] consider only the inflow forcing, assuming the momentum deficit and wake center to advect as passive tracers ~~in~~ a background flow. Typically, this effect is considered by applying a low-pass filter to the turbulent inflow such that only structures that exceed a certain minimum length scale influence wake motion. This cut-off length scale is often set to twice the observed instantaneous wake diameter, or simply as twice the rotor diameter ($2D$) in most impletementatinos.[39,40] By combining advanced measurement techniques with sophisticated data analysis methods, we seek to develop an understanding of wake meandering across a wide range of atmospheric conditions. We relate and its impact on wind farm performance, ultimately contributing to the optimization of wind energy production and turbine longevity.

II. METHODS

A. Wake meandering

Wake meandering is quantified in this work through the dynamics of the transverse coordinate of the wake center (μ), which we estimate by fitting a Gaussian profile to the momentum deficit derived from the lidar scans. The velocity deficit is defined as,

$$\tilde{u}(x, y, t) = 1 - \frac{u(x, y, t)}{U_{\text{hub}}(t)} \tag{1}$$

In Eq. (1), u represents the streamwise velocity measurement, estimated from the cosine of the line-of-sight velocity observed by the lidar. Because our data are collected in approximately horizontal planes at hub height, we only consider lateral movements of the wake center along the y-direction at a fixed height above ground $z = z_{hub}$. The wake center is detected through a least-squares fit of a Gaussian function to the observed U by the lidar at each downstream location and for each time.

$$\tilde{u}(x, y, t) = A \exp \frac{1}{2} \left[\frac{y - \mu(x, t)}{\sigma(x, t)} \right]^2 + C \tag{2}$$

In Eq. (2), the peak momentum deficit is denoted as A, the standard deviation corresponding to wake width is σ, and the lateral wake center is μ. An offset term, C, is included in the function, although it is typically a relatively small value, $-1 \lesssim C \lesssim 1$.

B. Snapshot Proper Orthogonal Decomposition

Proper Orthogonal Decomposition (POD), also known as Principal Component Analysis ~~(PCA)~~ or Karhunen-Loève decomposition, has been widely used in fluid dynamics for identifying coherent structures and reducing the dimensionality of complex flow fields[31]. However, its application to data in polar coordinates, as is more natural for scanning lidar observations, presents unique challenges that necessitate a careful reconsideration of the method's foundational elements.

Snapshot POD begins with a sequence of lidar observations $\mathbf{X} = [\mathbf{x}_1,\ ...,\ \mathbf{x}_m]$ where each snapshot is a realization in the state space $\mathbf{x}_k \in \mathbb{R}^n$ at time t_k. POD seeks an orthonormal basis that optimally represents the data in terms of captured variance. When dealing with data in polar coordinates, the standard Euclidean inner product implicit in this formulation becomes inappropriate to adquately represent the distribution of tublent kinetic energy in the domain. The varying cell sizes in a polar grid lead to an inherent bias, overemphasizing contributions from larger radii. To address this, we modify the standard POD algorithm to incorporate a physically appropriate inner product for polar coordinates. Considering two snapshots $\mathbf{x}(r,\theta)$ and $\mathbf{y}(r,\theta)$, each of which represents the streamwise velocity (system state) vector in polar coordinates, we define a weighted inner product as,

$$\langle \mathbf{x}, \mathbf{y} \rangle_W = \int_0^{2\pi} \int_0^R \mathbf{x}(r,\theta) \cdot \mathbf{y}(r,\theta)\, r\, dr\, d\theta \tag{3}$$

For discrete data on a polar grid with spacing Δr and $\Delta\theta$ in the radial and azimuthal directions, resepectively, Eq. (3) becomes,

$$\langle \mathbf{x}, \mathbf{y} \rangle_W = \sum_{i=1}^{N_r} \sum_{j=1}^{N_\theta} \mathbf{x}(r_i,\theta_j) \cdot \mathbf{y}(r_i,\theta_j)\, r_i\, \Delta r\, \Delta\theta \tag{4}$$

The inner product in Eq. (4) can be represented by a diagonal weight matrix $\mathbf{W}$:

$$\mathbf{W} = \mathrm{diag}(r_1, ..., r_1, r_2, ..., r_2, r_3, ..., r_{N_r}, ..., r_{N_r})\Delta r \Delta\theta \tag{5}$$

where each radial coordinate, r_i, is repeated N_θ times.

The new inner product is incorporated into the POD algorithm by modifying correlation matrix to account for the weights described by Eq. (5) as,

$$\mathbf{C} = \frac{1}{m}\mathbf{X}'\mathbf{W}\mathbf{X} \tag{6}$$

The POD modes are then obtained by solving the eigenvalue problem in the standard fashion,

$$\mathbf{C}\mathbf{v}_i = \lambda_i \mathbf{v}_i \tag{7}$$

where λ_i are the eigenvalues and $\mathbf{v}_i$ are the corresponding eigenvectors. The POD modes ϕ_i are then reconstructed as:

$$\phi_i = \frac{1}{\sqrt{\lambda_i}}\mathbf{X}\mathbf{v}_i \tag{8}$$

This formulation ensures that the POD modes accurately reflect the true spatial scales and turbulent kinetic energy of the flow structures in polar coordinates, without bias towards structures at larger radii.

This adaptation of the standard snapshot POD algorithm maintains the core principles of POD while adapting it to the geometric constraints of polar coordinate systems, aligning with the theoretical perspective advocated by Holmes[41], that the choice of inner product in modal decompositions should reflect the physics of the problem at hand. By incorporating an inner product motivated by the polar description of the state vector, we extend the applicability of POD to the natural coordiante system of scanning Doppler lidars, and open the method to a wider class of atmospheric science and wind energy problems.

C. Reconstruction

The approach to velocity field reconstruction used in this work relies on the nature of the POD to organize input system dynamics into coherent structures. We develop a combinatorial approach to reduced order modeling to identify the turbulent structure (POD modes, ϕ_i) that contribute most to wake meandering. Each combination of modes was determined by selecting k of the first N_r modes, where $k \in [3, ..., 15]$ and $N_r = 15$ was determined by requiring at least 90% of the TKE represented in the observational basis for each case. Flow fields reconstructed with each unique combination of modes are then used to estimate wake meandering identically as for the lidar scans, as described in Section II A.

In each of the ROMs, the zeroth mode, ϕ_0, representing the mean flow, is also included. The total number of combinations for each dataset was $K = \sum_{k=3}^{14} \binom{N_r}{k} = 16,278$. Fig. 1 shows the number of combinations tested for each value of k.

For each ROM, velocity fields are reconstruction from the truncated basis, summing over selected mode indices, rather than a sequential set up to a maximum. For each combination $I = \binom{N}{k}$, the reconstructed velocity field is,

$$\hat{u}(\boldsymbol{x}, t) = \sum_{i \in I} a_i(t) \Phi^{(i)}(\boldsymbol{x}) \tag{9}$$

where the index i may take only the values included in a particular combination of modes and the hat notation indicates a low-dimensional representation of the velocity field. In Eq. (9), the velocity field and modal basis have been denoted as scalars, as only the streamwise

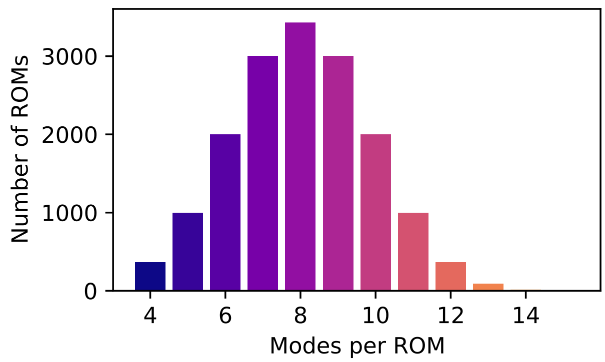

FIG. 1: 1 number of mode combinations used to reconstruct the flow field for a given maximum number of modes considered (k). For all cases, only modes 1 through 15 are considered ($N_r = 14$).

component of velocity is considered.

III. DATA

This study uses nacelle-mounted scanning lidar data and atmospheric inflow measurements from two recent DOE-funded field campaigns. The American WAKE ExperimeNt (AWAKEN)[42] is a multi-institutional campaign that gathers data on wind farm-atmosphere interactions, which contribute to uncertainty in wind plant performance models. Atmospheric conditions were characterized using data from a surface flux station and a meteorological tower at site A1.

The surface flux station, equipped with a Gill R3-50 sonic anemometer mounted on a 4-meter tripod, recorded data continuously at 20 Hz. Data acquisition software, originally developed by Argonne National Laboratory for the ARM ECOR system in 2003, was adapted for AWAKEN to work with the Gill Sonic R3 series anemometers[43]. Sensible heat flux and momentum flux were processed into 30-minute resolution data, assuming a constant relative humidity of 50% due to the absence of real-time humidity measurements. Hub-height wind speed and direction were recorded with a Thies 3D Ultrasonic anemometer. Turbulence intensity was calculated as the standard deviation of wind speed divided by the mean wind speed over a 10-minute rolling period. Wind directions were limited to 170° to 200°, corresponding to the peak of the wind rose in Fig. 2, which also restricted times when the wake from a neighboring turbine was visible.

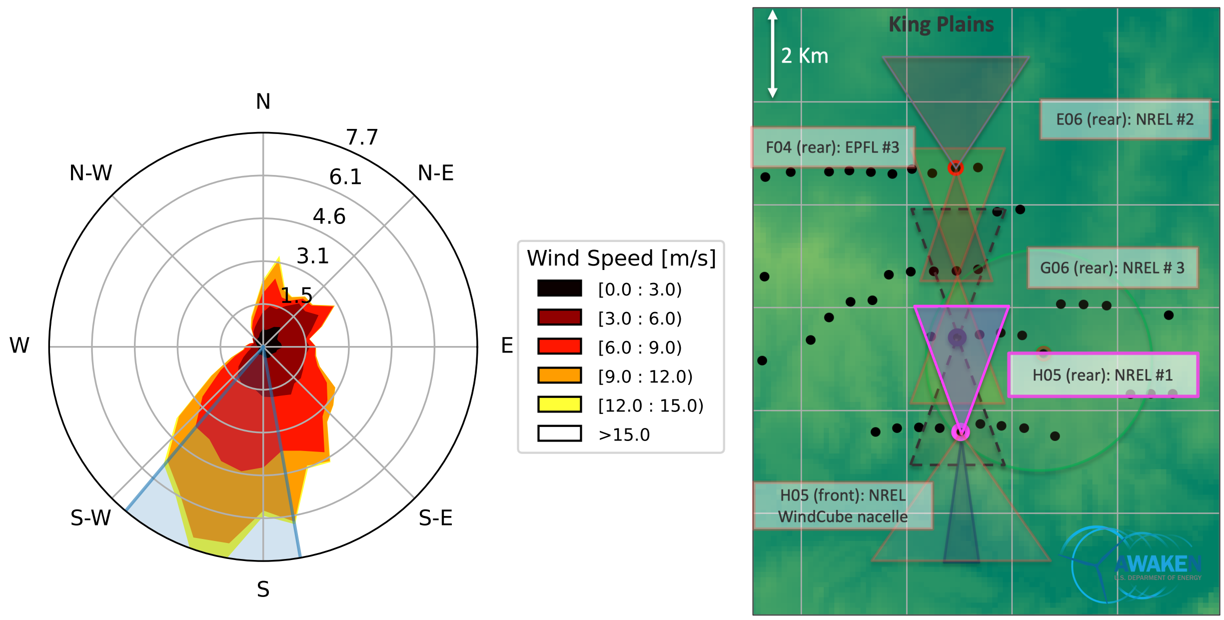

FIG. 2: Wind rose showing selected wind direction sector in blue (left) and high-level schematic (right) of the AWAKEN project. **add site A1, annotate for met tower, flux station**

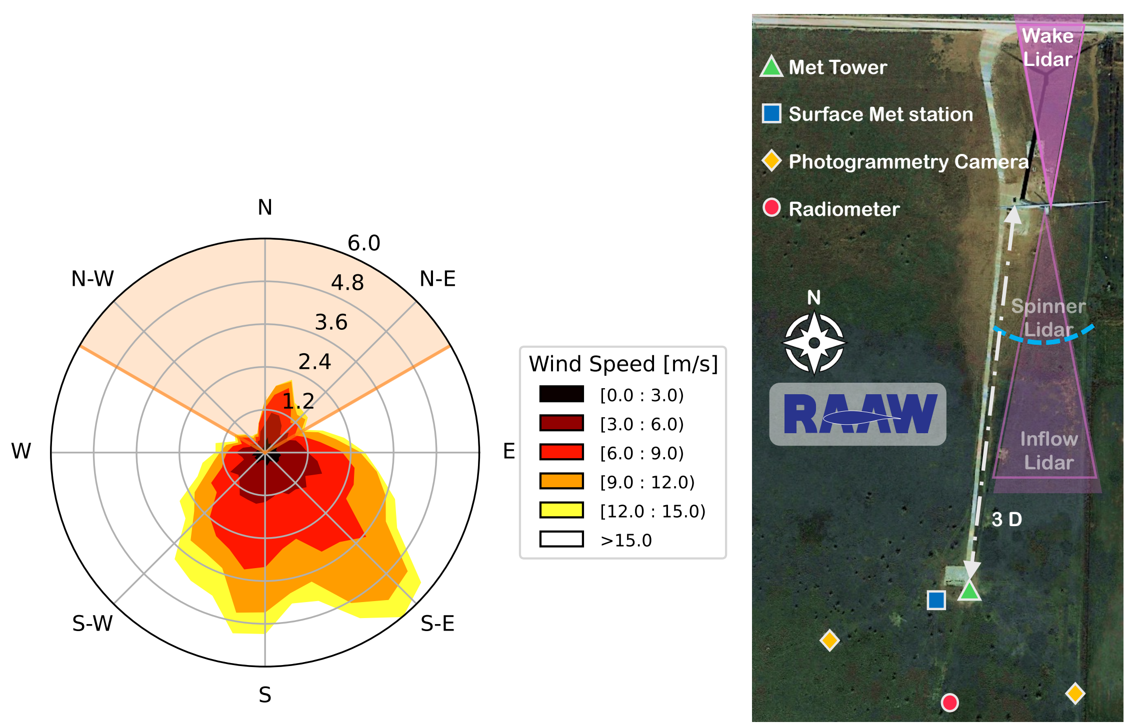

FIG. 3: Wind rose showing excluded wind direction sector in orange (left) and high-level schematic (right) of the RAAW project.

The Rotor Aerodynamics, Aeroelastics, and Wake (RAAW) project studies the performance of modern flexible rotor wind turbines. Atmospheric inflow conditions for the RAAW data cases are derived from a 183.5-meter guyed meteorological tower, instrumented with 5 cup anemometers, 3 ultrasonic anemometers, 3 vanes, 3 barometric and temperature sensors, and one humidity sensor. Ultrasonic anemometers reported data at 20 Hz, and all other instruments at 1 Hz. Near-surface measurements of momentum and heat flux were provided by a surface met station, deployed 10 meters southwest of the meteorological tower. This station hosted an ultrasonic anemometer 2.5 m above ground and probes for atmospheric pressure, temperature, and humidity 2 m above ground. As with the tall tower sensors, the anemometer reported data at 20 Hz, and the other sensors at 1 Hz. Figure 3 shows the experimental arrangement of the RAAW project, including the met tower, surface met station, and turbine. **NEED A RAAW CITATION, when will the report be done?**

Both the RAAW and AWAKEN projects feature GE 2.8 MW turbines with 127 m rotor diameters. The RAAW project used a prototype R&D turbine in Lubbock, TX, with a hub height of 120 m, while AWAKEN used production-run GE2.8-127 turbines with a hub height of 88.5 m. Both projects employed Halo Photonics Streamline XR+ Doppler scanning lidars positioned on the aft sections of the nacelles. These were configured for various measurement strategies, including plan-position indicator (PPI) scans to measure fluctuating wake flow fields.

Wake PPIs were designed using the LiSBOA tool[26], sweeping approximately 1° of azimuth per second, covering a sector of approximately 20° centered downstream of the turbine. **Verify scan rates and limits.** Lidar data were quality controlled using the Field EXperiments Tool Arsenal (FIEXTA) software[44], which implements dynamic filtering methods from Beck and Kühn[45]. Data were temporally upsampled using local advection velocities and the time delays between successive beams at similar azimuth angles[46], producing line-of-sight velocity 'snapshots' at a temporal resolution of 1 second.

TABLE I: **build table describing data, lidar scan parameters for AWAKEN, RAAW**

Scanning lidar provides line-of-sight wind speed, v_{los}. The horizontal wind speed, u was retrieved as,

$$v_{\text{los}}(r) = u \cos\zeta \cos(\psi - \theta) + w \sin\zeta \tag{10}$$

where u is the horizontal wind speed, r is the range along the laser beam, ζ is the elevation angle, ψ is the azimuth angle, and θ is the wind direction. For all lidar measurements used in this analysis, the elevation angle was fixed at 0°, simplifying the relationship between v_{los} and u.

While the lidars are capable of making measurements up to 4 km from the wind turbine, the measurement qualilty is typically quite low, and wakes are often fully recovered far sooner. Additionally, the data collected in the AWAKEN project include only inflows from a southerly sector, and the downstream row of turbines is often visible in the scan data. To preclude observations of wakes of neighboring turbines from influencing wake center estimates, which would complicate the modal analysis undertaken below, scans are constrained to ranges where $2.5 \leq x/D \leq 10$ and azimuth angles $-20^\circ \leq \alpha \leq 20^\circ$. **check azimuth limits**

Each data set of continuous PPI scans lasted approximately 30 minute, with a full azimuthal sweep occuring approximately every 18 to 20 seconds. After spatiotemporal upsampling ~~following the procedure outlined in Beck and Kühn[46]~~, each data case from the AWAKEN project contained approximately 1,650 snapshots (complete, infilled scans) at temporal resolution of 1 s. Data cases from the RAAW project are contained in 15 minute sections, leading to upsampled and infilled data sets of approximately 800 scans each at a temporal resolution of 1 s.

Measurement cases combining lidar data sets and inflow measurements were considered only after passing through the following quality control filters. Each filter is followed by a metric indcating the portion of the population of data that were disallowed and a description. Note that more than one filter condition was applicable to many of the data points.

Sensor Quality—6%: Measurement cases combining lidar data sets and inflow measurements were considered only after filtering out stuck, broken, or defective sensors. This ensures that all required measurements are reliable and eliminates cases where data are missing or sensors fail to report reasonable observations.

Wind Direction Sector—34%: Cases were filtered to include only wind directions within the appropriate sector. This is crucial because we rely on the characteristics of the inflow to contextualize wind turbine wake meandering. For the AWAKEN project, this step also ensures that only the wake dynamics from the single turbine of interest are present in the lidar scans.

Yaw Travel Limitation—51%: Only cases with a total yaw travel below 20° were considered. In both the RAAW and AWAKEN projects, the wind turbine operates at a nominal control point, allowing the rotor to yaw and optimally align with the incoming wind. Yaw activity is evident in lidar scans as abrupt changes or sweeping movements of the wakes within the scanned sectors. Very few lidar scan periods are completely free from yaw activity, hence the 20° travel limitation.

Power Production Threshold—18%: Cases were filtered to include only those where active power is greater than 150 kW and that power generation is within one standar deviation of the nominal power curve. This criterion ensures that the turbine is operating nominally and that a wake will be evident in the lidar data. The selection allows for describing wake meandering under a wide range of atmospheric conditions and inflow wind speeds.

Variability Filtering—10%: As a final quality control step, cases were excluded where the total variation exceeded the 90$^{\text{th}}$ percentile. Following the method outlined by Hamilton,[47] total variation is defined as the determinate of the covariance matrix relating inflow wind speed, wind direction, turbulence intensity, turbine power production, and nacelle position signals from the SCADA record. This step ensures that only statistically stationary conditions are considered, eliminating cases with drastically changing atmospheric conditions or significant turbine control actions during the lidar scan period.

A total of 65 cases were retained after applying each of the quality control filters enumated above, including 54 cases from the AWAKEN proejct and 11 cases from the RAAW project. Figure 4 shows the distribution of the full range of cases with wake PPI scans. Cases that did not pass all of the quality control measures listed above are shown in gray. The final cases selected for analysis are highlighted in the figure according to the total yaw travel observed during the respective wake scanning period.

The integral time scale of atmospheric turbulence provides a measure of the temporal

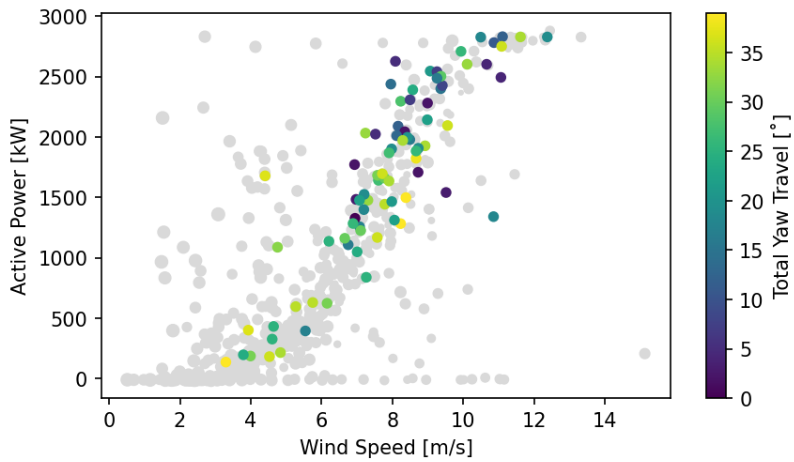

FIG. 4: Power curve from turbine H05 at the King Plains Wind Plant. All available cases are shown with gray points, cases selected for analysis in the current work are colored by the total yaw travel of the turbine during the measurement period. **will this figure make it to the final version?**

correlation of velocity fluctuations in the atmospheric boundary layer. The autocorrelation coefficient $\rho(\tau)$ is then defined as:

$$\rho(\tau) = \frac{\langle u'(t)u'(t+\tau)\rangle}{\sigma^2} \tag{11}$$

where $u'(t)$ represents the turbulent velocity fluctuations, τ is the time lag, $\langle\cdot\rangle$ denotes the ensemble average, and σ^2 is the variance of the velocity fluctuations. The integral time scale T is calculated by integrating the autocorrelation function up to the point where it decays to a specified threshold:

$$T = \int_0^{\tau_c} \rho(\tau), d\tau \tag{12}$$

Here, τ_c represents the correlation decay limit, typically chosen as the point where $\rho(\tau) = 0.05$, rather than the first zero-crossing point.[48,49] This integration captures the temporal coherence of turbulent fluctuations, providing insight into the characteristic time scales of atmospheric turbulence. In practice, this calculation was performed using 20-Hz wind speed measurements from a sonic anemometer positioned at hub height for the turbines in each project, with computations based on 30-minute averaging periods.

Figure 6 shows the distribution of inflow wind speed, U_∞, integral time scale, T, and inflow Strouhal number, St_{in} for each of the 71 cases from the combined RAAW aand AWAKEN experiments. The integral time scales, T, are taken as the characteristic period of the fluctuation in the velocity field and are used to define the characteristic frequency of turbulent fluctuations in the discussion of meandering frequencies.

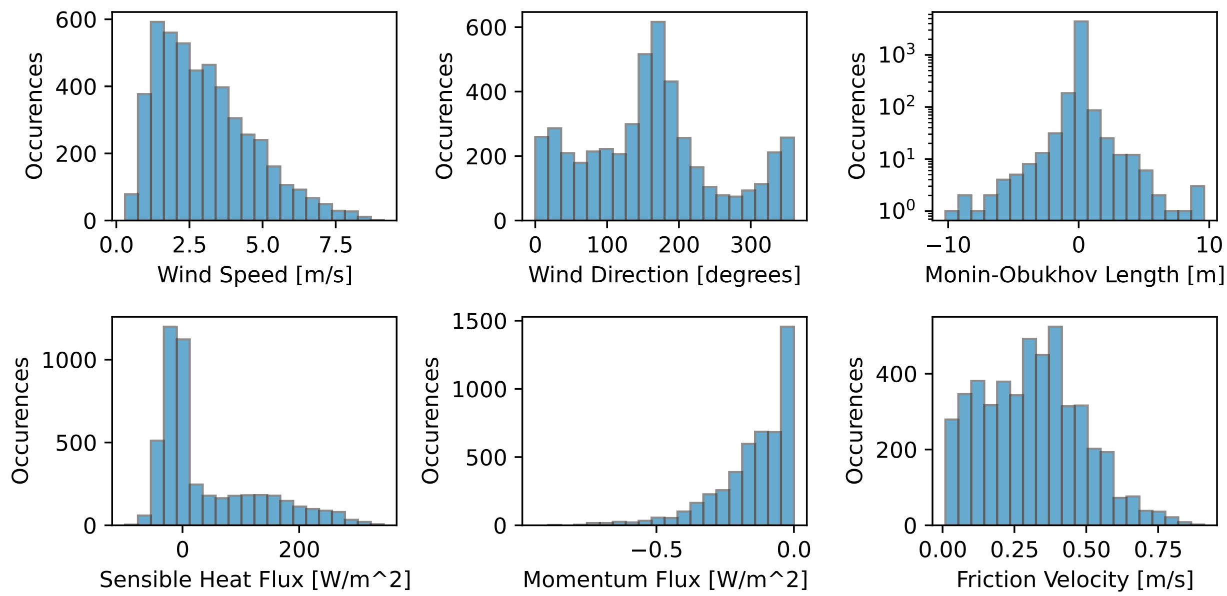

FIG. 5: Distributions of the average inflow conditions observed at the surface met stations at Site A1 throughout the AWAKEN and RAAW campaigns.

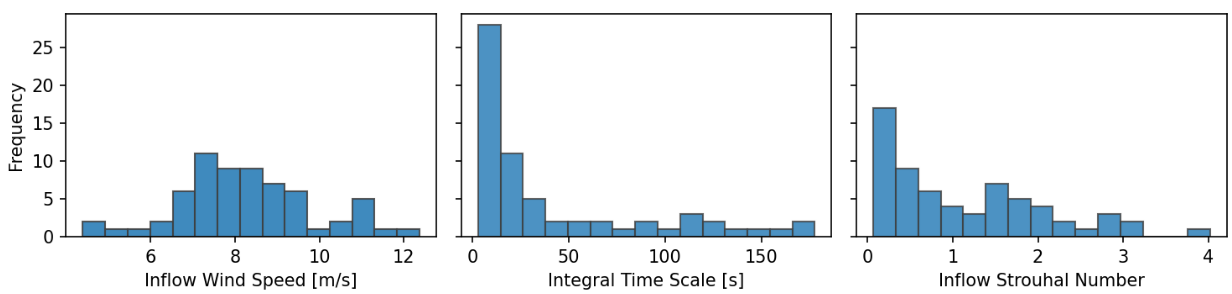

FIG. 6: Distributions of the average inflow wind speed, U_∞, integral time scale, T, and Strouhal Number, St_{in}, at hub height for the subset of conditions corresponding to the 65 selected cases.

IV. RESULTS

Average scans shown for selected cases are shown in Fig. 7 and highlight differences in the peak momentum deficit and recovery rate of wakes in different atmospheric conditions.

While all data sets include some yaw activity, the average wake is centered around $\alpha = 0$. Each of time-averaged velocity fields in Fig. 7 highlights characteristics of the wind turbine wake as well as the influence of the atmsopheric forcing conditions and effects of the wind turbine operations and controls. The data quality control preferntially selecetd the most statistically stationary cases from the full population of data. As a consequence of this selection, there is a bias toward stable atmospheric conditions in which the yaw travel was minimized during the lidar collection periods of both experiments.

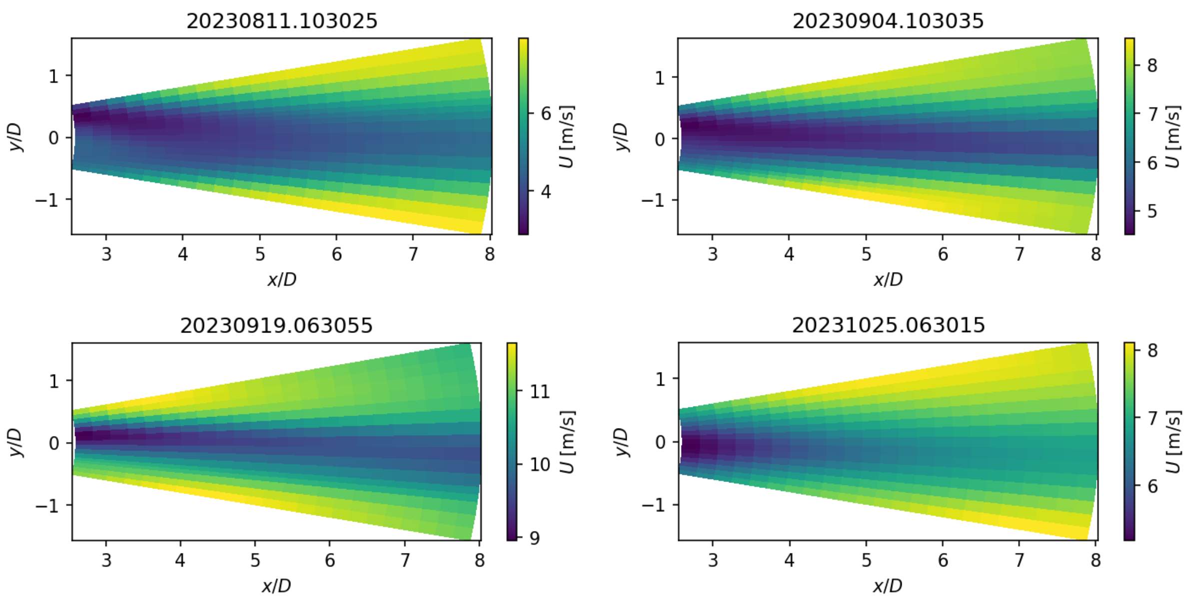

FIG. 7: Average wake velocity fields for 2023-08-11 10:30 (top left), 2023-09-04 10:30 (top right), 2023-09-19 06:30 (bottom left), 2023-10-25 06:30 (bottom right) showing a range of background velocities, momentum deficits and wake recovery rates.

Yaw activity is not directly evident in the time-averaged wakes. However, Fig. 8 shows a time-azimuth cross section of a single dataset at a range of approximately $x/D = 8$ (1 km), in which the wake is highlighted as a low wind speed region that persists in time across the figure. Meander in the azimuthal direction seen in the figure combines the effects of several factors. While the wake center does travel horizontally, there are also translations of the wake center introduced by the yaw activity of the rotor. Horizontal wake center locations, μ are estimated as the center loaction of the Gaussian function fit to the velocity deficit, as in Eq. (2). Yaw activity becomes especially evident in the time histories of detected wake centers, presented in Fig. 9. Yaw activity, illustrated by the dashed black line and related

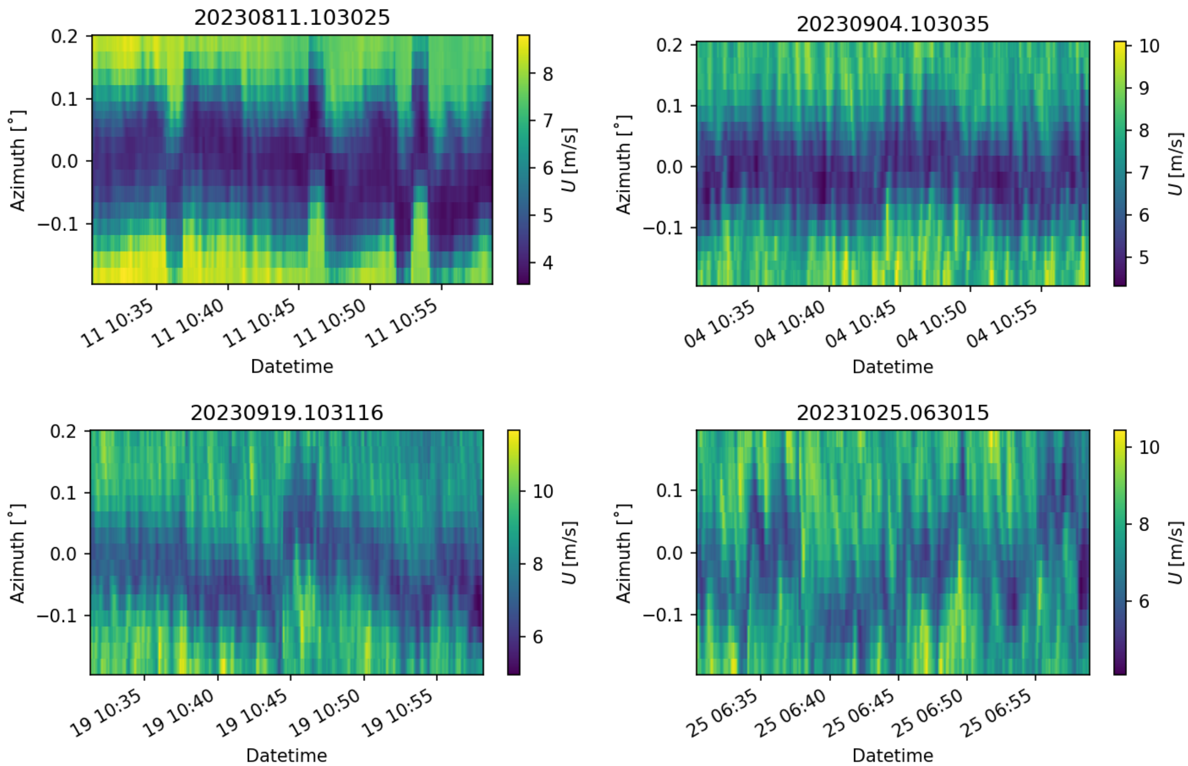

FIG. 8: Average wake velocity fields for 2023-08-11 10:30 (top left), 2023-09-04 10:30 (top right), 2023-09-19 06:30 (bottom left), 2023-10-25 06:30 (bottom right) showing a range of background velocities, momentum deficits and wake recovery rates.

vertical axis on the right, are evident in the detected centers as sudden shifts in the wake center time histories. Because changes in the nacelle position impact the apparent wake centers uniformly at all ranges, a simple correction can be applied to the wake centers as, $\Delta\mu = r \sin\psi$, where r indicates the distance from the lidar (range) and ψ indicates the nacelle orientation with respect to the position at the beginning of each data set. Correcting for changes in nacelle position throughout each dataset is important to correctly characterize the distributions of wake centers and their dynamics in the turbine frame of reference. It should be noted that turbines take yaw-correcting actions only after after an integrator of error between the measured wind direction and the nacelle position exceeds a threshold. This means that in most cases there remains some influence of yaw misalignemnt on the detected wake centers even after correcting for yaw activity. Without a time history, it is not possible to accurately account for the yaw error. However, this sort of misalignment is part of the nominal operation of a turbine under normal control conditions, and is a possible trigger for wake meandering.

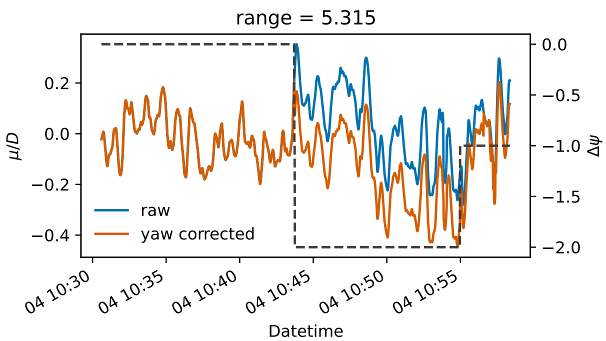

FIG. 9: Detected wake centers directly from lidar observations (blue) and after correcting for yaw activity of the turbine (orange). The dashed black line indicates nacelle position changes from the beginning of the time series and corresponds with the axis labels on the right.

An example of the momentum deficit measured by the lidar is shown in Figure 10 in blue, along with local fits of the Gaussian function in black and detected wake centers as red points. Each of the subfigures indicate a good level of agreement between the momentum deficit and fit function, building confidence in the Gaussian model of the wake. Alternative methods of detecting the wake centers have been explored in the literature,[2] including a momentum-centroid approach and local minimum velocity methods, but are not explored in the current work for brevity.

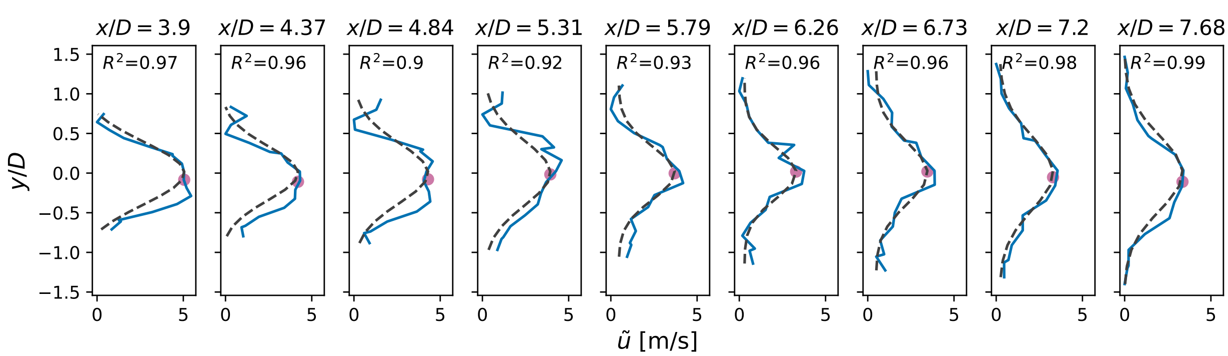

FIG. 10: Detected wake centers showing Guassian fit against momentum deficit. Blue lines indicate momentum deficit estimated from lidar scans, dashed line shows the fit Gaussian profile, point marker is is the detected wake center μ/D. ~~add R^2 value to each supblot~~

An alternative view of the detected wake centers is provided in Fig. 11, where μ/D is noted on the lidar scans as points colored by their respective R^2 values. In both examples, it is evident that the μ/D follows the bluk momentum deficit in the wake quite closely. A bias of μ/D toward positive values of y/D in the near wake $x/D \lesssim 3$ is also visible in the scan taken at 2023-09-04 10:42:15 UTC (left). This bias, which persists throughout many of the scans, arises from the asymmetric double-Gaussian profile of the momentum deficit very near the turbine.[50–52] Positive bias of μ/D in the near wake is not easily corrected, and leads to impacts later in the analysis, including power spectral densities an representation with the ROMs.

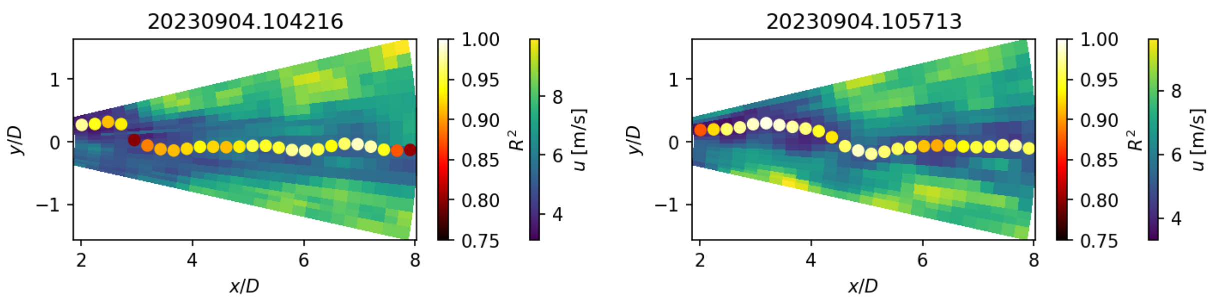

FIG. 11: Example lidar scans showing u overlaid with the fitted wake centers. Detected wake centers are colored by their respective coefficent of determination.

Wake meandering combines the downstream advection of μ by the mean flow and turbulent structures in the wake. Trajectories of μ throuhgout the measurement domain are shown in Fig. 12 (left) for several times in the data set. ~~The trajectories all show the positive bias of μ in the near wake and then tend toward negative values of y as the range (downstream distance) increases.~~ The meander is also evident in the trajectories as a local tendency back toward positive values of y. The amplitdues of these meandering structures also tend to increase as they advect downstream. Kernel density estimates of wake center locations are shown in Fig. 12 (right), highlighting the tendency of the μ to spread out at greater streamwise distances.

In addition to statistical descriptions of the wake center, we are also interested in dynamics of wake meandering, specifically relating the characteristic frequencies of wake meandering, quantified through power spectral densities (PSD, Eq. (13)) to inflow conditions and the POD modes, ϕ_i. PSDs are calculated using Welch's method as,

$$S_{\mu\mu}(f) = \frac{1}{L}\sum_{k=1}^{K}\frac{1}{N}\left|\sum_{n=1}^{N}\mu_k(n)e^{-i2\pi fn/N}\right|^2 \tag{13}$$

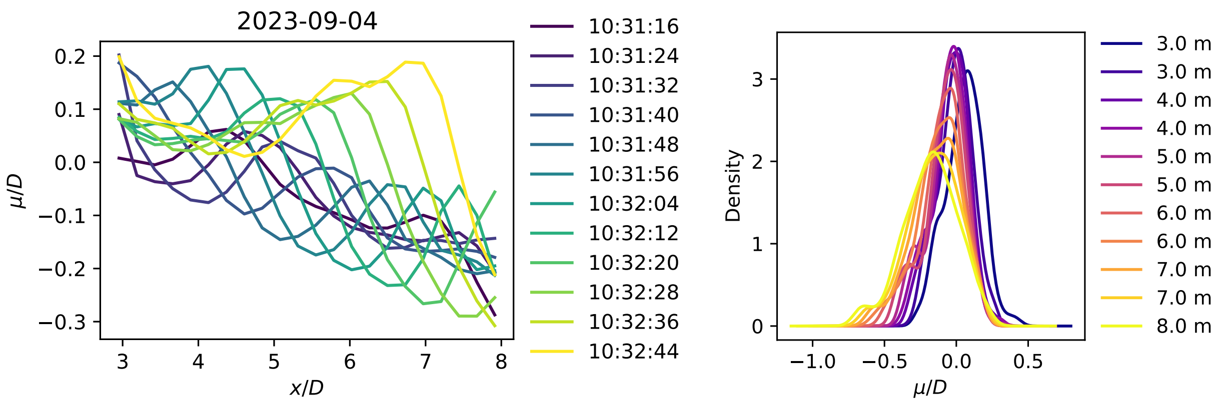

FIG. 12: Wake center trajectories for selected times (left) and kernal density estiamtes of wake centers by distance downstream of the lidar (right).

where, K is the number of segments, $L = 512$ samples is the number of points in the FFT $N = 1,024$ is the length of each segment, $\mu_k(n)$ is the k-th segment of the time series. A uniform window function is applied to each segment in the calculation of the PSDs. The frequency range for all spectra in the current work have been normalized with the average inflow velocity recorded by the met towers, U_∞, and the turbine's rotor diameter, D, as $f_{St} = f * D/U_\infty$

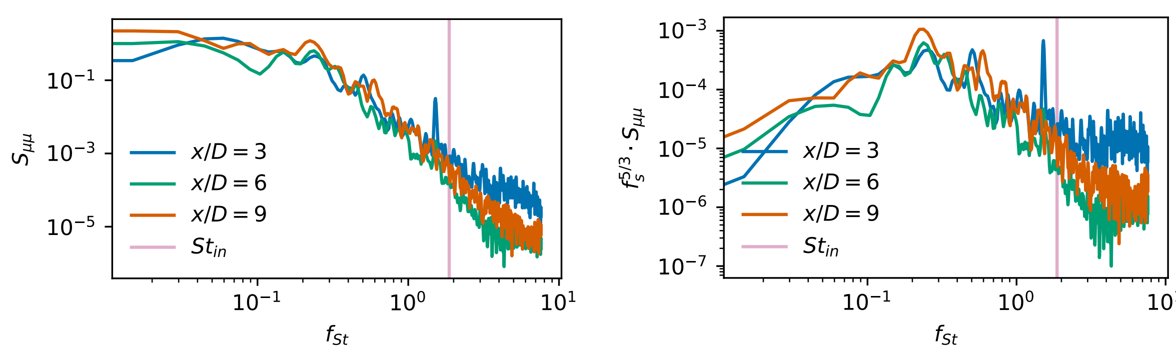

FIG. 13: PSDs (left) and premultiplied spectra (right) of detected wake centers from the lidar data at selected downstream distances. ~~fix vertical axis label for premult spectra~~

The snapshot POD introduced in Section II B requires state space observations with consistent dimensions and coordinates, making it impractical to correct for yaw before decomposition. Remapping wake observations from lidar scans onto a corrected reference frame downstream of the turbine would introduce inconsistencies in the azimuthal coordinates or create empty radial slices, both of which would complicate the eigenvalue decomposition central to the POD analysis.

Each of the 65 cases have azimuth coordinates that vary slightly due to the continuous scan operation of the lidar, which assigns nominal azimuth angles after a set time delta. Operational delays, data transmission, and status checks cause small variations in the number of scans per dataset. This analysis focuses on the low-rank POD modes that account for most of the turbulent kinetic energy (TKE) in the lidar scans. Although the energy distribution across POD modes varies between datasets, researchers order the modes by descending energy, with the eigenvalue λ_i of each mode representing the energy distribution

We use only the first 14 POD modes, which capture the most coherent structures in the lidar scans, to reconstruct flow fields for meandering analysis. The eigenvalues λ_i quantify the TKE captured in each reduced-order model (ROM), directly measuring reconstruction error. Missing energy in a ROM remains bounded by the smallest mode eigenvalue included in the reconstruction. Figure 14 illustrates the distribution of TKE described by the POD modes across all cases. This analysis excludes the energy of the zeroth mode ϕ_0, which represents the mean flow field, ensuring the normalized sum represents only the turbulent kinetic energy. The black dashed line in the figure identifies the threshold at which the POD modes capture 90% of the TKE. In some cases, as few as 4 modes meet this threshold, while other cases require more modes. In all but two outlier cases, the POD meets the 90% TKE threshold with a maximum of 14 modes.

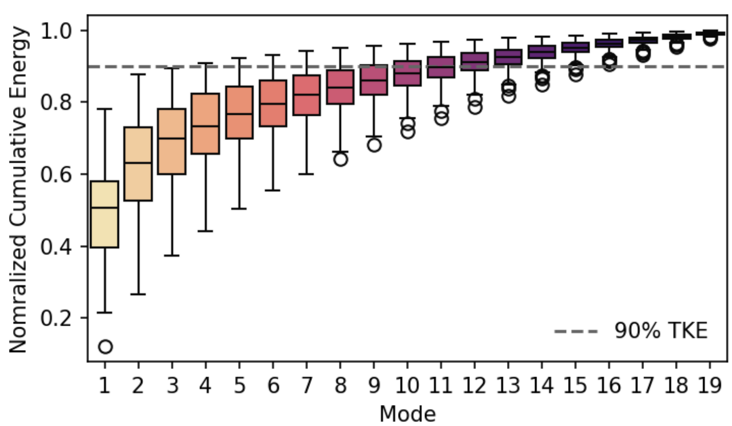

FIG. 14: The distribution of cumulative summations of eigenvalues associated with each case. Including

We apply the POD independently to each lidar dataset, producing a unique basis of modes for each case. Figure 15 shows the POD modes associated with the case beginning at 2023-09-04 10:30 UTC. Although the mode bases differ quantitatively for each case, their structure and order remain qualitatively similar. Researchers include colorbars with each mode for completeness, although the units and quantitative values only become meaningful when ~~they combine the modes with~~ their respective coefficients to form a ROM.

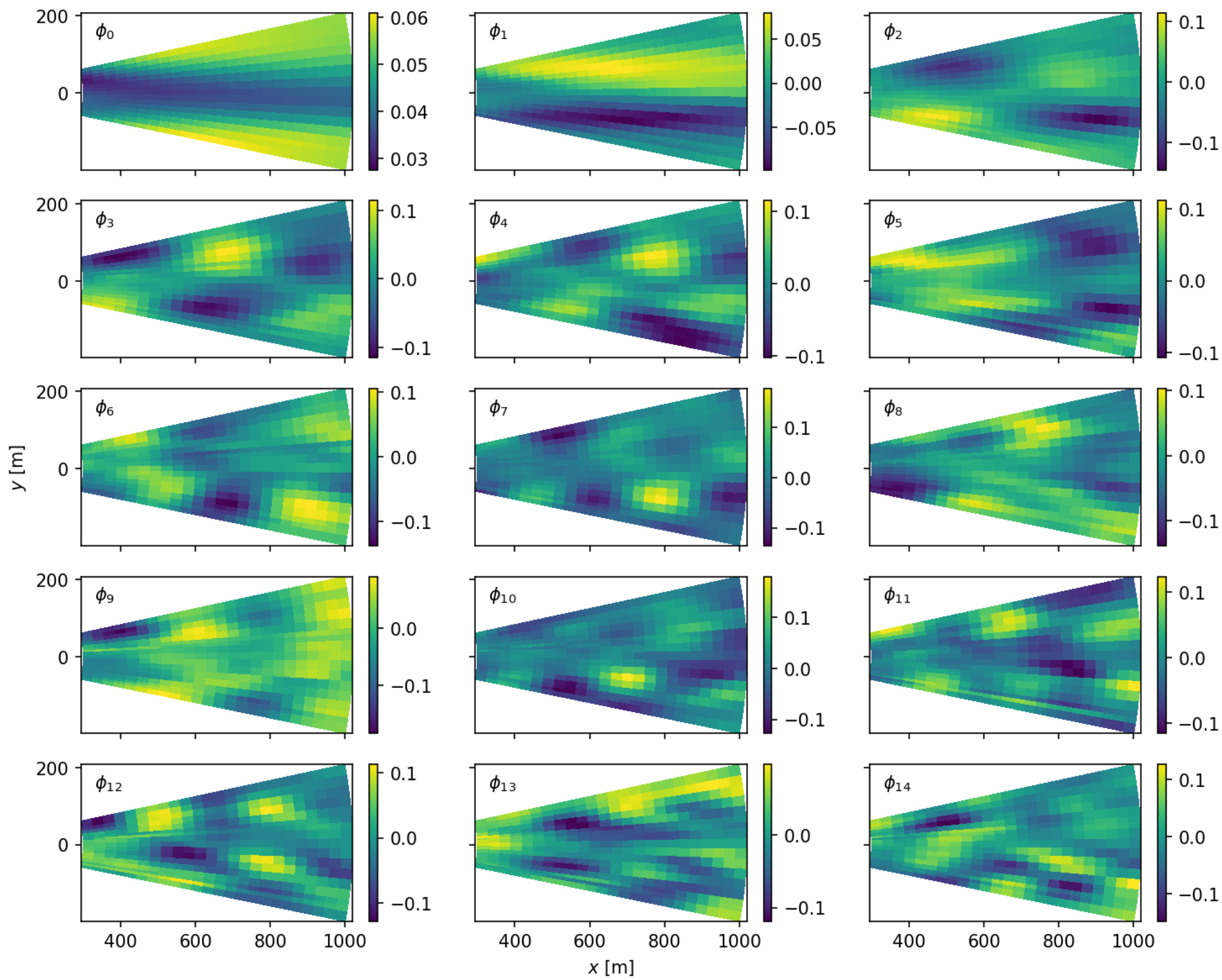

FIG. 15: POD modes from the example case on 2023-09-04 10:30 UTC.

The zeroth mode, ϕ_0, represents the mean flow, capturing the characteristic momentum deficit in the wind turbine wake (see Table ??). Modes ϕ_1 through ϕ_{14} describe coherent wake structures ranked by their contributions to the lidar scans. The first mode, ϕ_1, accounts for 20% to 8% of the turbulent kinetic energy (TKE) and primarily describes lateral shifts in the wake, crucial for wake meandering. Mode ϕ_2 represents a full oscillation in the streamwise

TABLE II: Description of POD mode structure, periodicity, and associated eigenvalue.

Mode, i	Structure	Eigenvalue, λ_i	Description
0	symmetric		Mean momentum deficit of the wake
1	antisymmetric		Full period in y, half-period in x.
2	antisymmetric		Contains a full period each in x and y
3	antisymmetric		Represents 1.5 periods in x and one period in y
4	asymmetric		Significant azimuthal and radial asymmetry
5	symmetric		Describes 1 full period in x
6	symmetric		1.5 periods in x
7	asymmetric		Highlights shear layer at negative y
8	symmetric		May represent 1.5 periods each in x and y
9	symmetric		Describes 2 full periods in x
10	asymmetric		Phase-offset, but similar to phi_7
11	asymmetric		Highlights shear layer at positive y
12	symmetric		Shows 2.5 periods in x, 2 periods in y
13	symmetric		Phase-offset, but similar to phi_{12}
14	asymmetric		Higher order, but similar to phi_7

(x) direction and a half oscillation in the transverse (y) direction, reflecting large streamwise wake variations. Mode ϕ_3 introduces an additional half-period in x, describing 1.5 periods in the lidar scan domain.

Higher-order modes become increasingly complex and difficult to assign a consise description. Mode ϕ_4 exhibits significant azimuthal asymmetry, complicating direct interpretation. In contrast, ϕ_5 is symmetric, describing a full oscillation in x and possibly reflecting vertical wake meandering. Mode ϕ_9 describes two full periods in x, further illustrating the complexity of higher-order structures.

Following Eq. (9), we reconstruct the flow field using 16,278 unique subsets including between 3 to 14 POD modes in addition to the mode descibing the mean flow (ϕ_0). Figure 16 shows the combinations, ranging from ROMs with only three turbulent modes and ϕ_0 on the ~~right,~~ to those incorporating more modes on the right. Each reconstructed wake profile was fitted to a Gaussian function (Equation 2) to estimate the lateral wake center, $\hat{\mu}$.

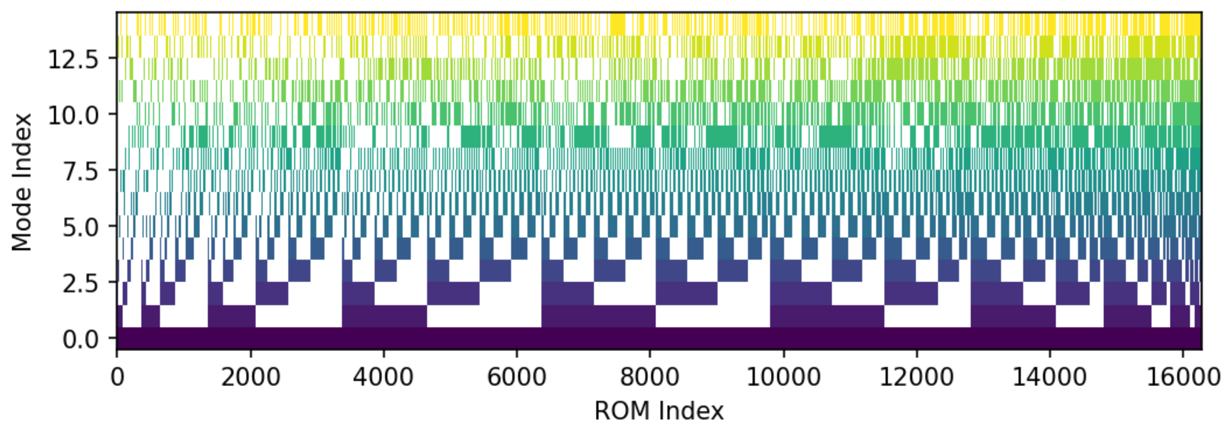

FIG. 16: Combination tree indicating which modes are contained in any given ROM. Modes are indicated by the colored regions, and their absence is shown in white. ~~change cbar label~~

When reconstructing velocity fields using POD modes, we aim to match the power spectral density (PSD) of wake center locations from the reconstructed field, $S_{\hat{\mu}\hat{\mu}}$, to the original field, $S_{\mu\mu}$. We consider mode combinations that reproduce wake meandering (i.e., $S_{\mu\mu} \approx S_{\hat{\mu}\hat{\mu}}$) representative of wake meandering in the low-order velocity field. Coherence, defined in Equation 14, quantifies the similarity between wake center trajectories in the lidar scans and ROMs:

$$C_{\mu\hat{\mu}}(f,x) = \frac{|S_{\mu\hat{\mu}}(f,x)|^2}{S_{\mu\mu}(f,x)S_{\hat{\mu}\hat{\mu}}(f,x)}. \tag{14}$$

Here, coherence $C_{\mu\hat{\mu}}(f,x)$ measures the similarity of wake center dynamics at frequency f. A value of $C_{\mu\hat{\mu}}(f,x) = 1$ indicates a perfect match between PSDs, while $C_{\mu\hat{\mu}}(f,x) = 0$ indicates no match.

PSDs of wake centers show better agreement at lower frequencies ($f_s < St_{in}$), which the POD modes in ROMs capture more accurately. Coherence decreases at higher frequencies, but because wake meandering is predominantly a low-frequency phenomenon, this reduction in coherence at higher f_s is acceptable for the purposes of this study. Figure 17 highlights the average coherence across the full frequency range for several cases. Good agreement between $S_{\mu\mu}$ and $S_{\hat{\mu}\hat{\mu}}$ is observed, particularly for $x/D > 4$. Highlighted regions in Figure 17 show where $f_s > St_{in}$, corresponding to reduced frequencies above the inflow Strouhal number, that are not typically associated with wake meandering.

Focusing on specific frequency ranges of $C_{\mu\hat{\mu}}$ provides a more accurate evaluation of each ROM's ability to reproduce wake meandering. High-frequency spectra are irrelevant for this analysis, as they do not reflect the physics of wake meandering. We evaluate the trends for ROMs in Figure 17 by calculating the average coherence, $\overline{C}_{\mu\hat{\mu}}$, over the respective frequency ranges.

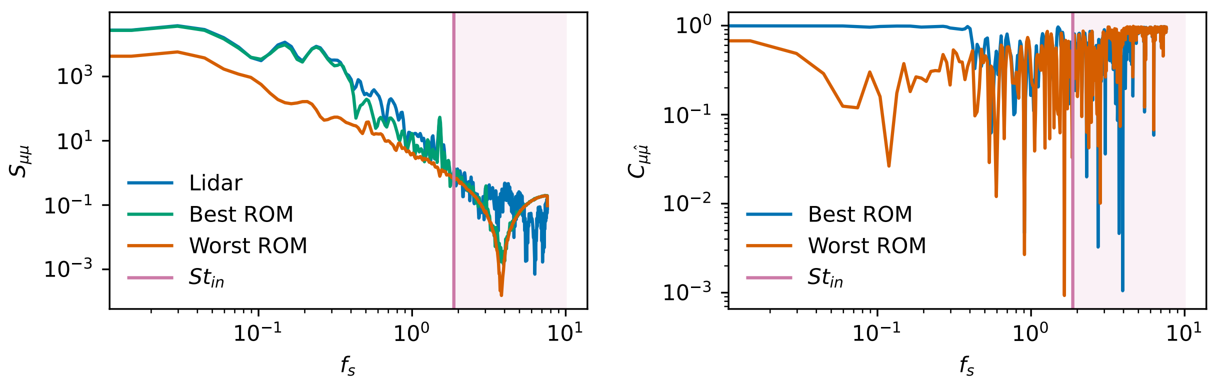

FIG. 17: Spectra (left) and coherence (right) for the ROMs that compare best (green) and worst (orange) to the PSD of μ from the lidar scans.

A more complete assessment of ROM quality appears in Fig. 18, which shows the average coherence over the reduced frequency, $\overline{C}\mu\hat{\mu}$. The top subfigure displays $\overline{C}_{\mu\hat{\mu}}$ for the full frequency range, with an average coherence of approximately 0.3 for most cases. The bottom subfigure, restricted to $f_s \leq St_{in}$, reveals more consistent results across blocks of ROMs. Referring to the mode distribution in each ROM shown in the combination tree (Fig. 16), a clear correlation emerges between $\overline{C}_{\mu\hat{\mu}}$ and the inclusion of specific modes.

Fig. 18 also reveals the dependence of $\overline{C}_{\mu\hat{\mu}}$ on the downstream distance from the lidar, as indicated by the vertical axes. By averaging along the range coordinate, we aggregate this information into a single scalar metric for each ROM, $\varepsilon = \langle \overline{C}_{\mu\hat{\mu}} \rangle$, where the angle brackets denote averaging along the streamwise coordinate. This allows us to describe the ROM distribution with a quantitative fit quality metric. Figure 19 (left) shows ε arranged by ROM index across the 16,278 mode combinations. The data in Fig. 19 reflect the same information as Fig. 18, averaged over the vertical axis (streamwise coordinate). On the right side of Fig. 19, the distribution of ROM quality is shown according to the number of modes used in each ROM. Although including more modes generally improves the representation of wake meandering, several exceptions to this trend exist. The distributions in Fig. 19 (right)

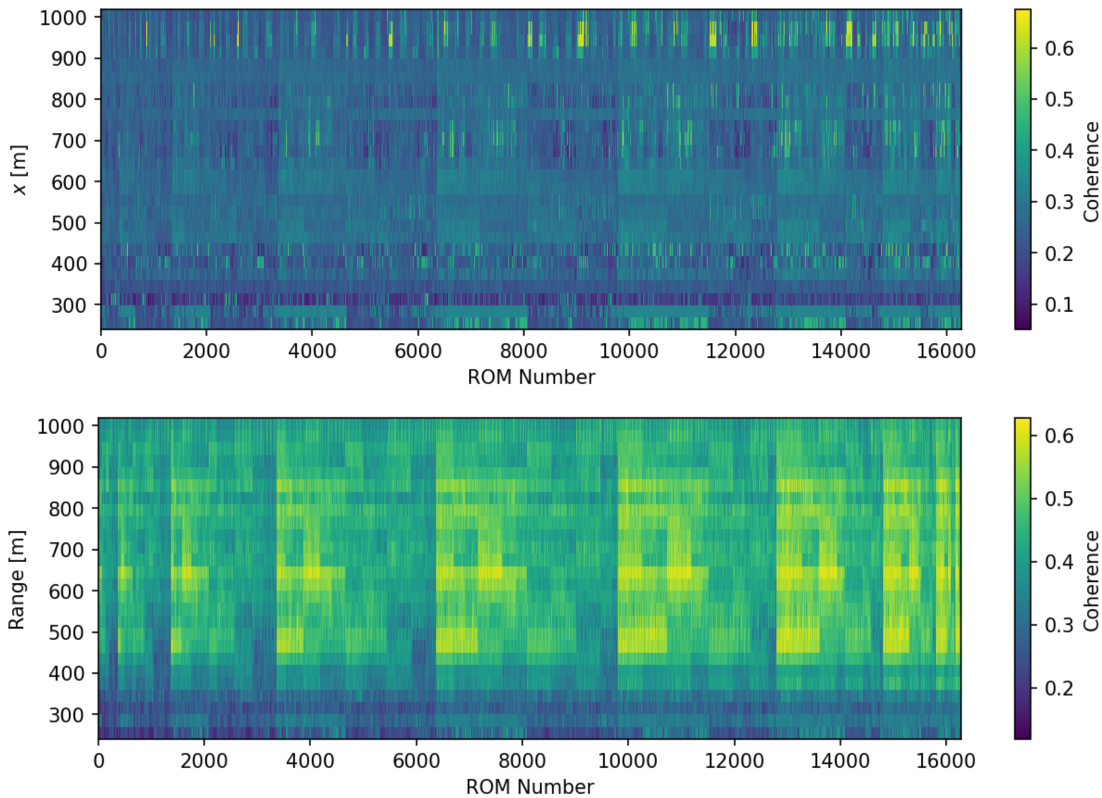

FIG. 18: Frequency-average coherence between detected wake centers from lidar scans and ROMs considering the full measured frequency range (top) and only frequencies $f_s \leq St_{in}$ Hz (bottom).

show that some ROMs with only four modes outperform others with 10 or more modes.

To investigate the sensitivity of wake center coherence to POD modes, we use an Analysis of Variance (ANOVA) approach.[53] We develop a statistical model to decompose the ~~observed~~ coherence ~~metric~~ into contributions from individual modes and their interactions. The model takes the following linear decomposition form, where ε is the average coherence, a quality metric for each ROM:

$$\varepsilon = \bar{\varepsilon} + \sum_{i=1}^{n} \beta_i q_i + \sum_{i<j}^{n} \gamma_{ij} q_i q_j + \xi \tag{15}$$

Here, $\bar{\varepsilon}$ represents the global mean coherence, β_i are the main effects of individual modes, and γ_{ij} capture the interaction effects between modes. The binary variables q_i indicate the presence (1) or absence (0) of each mode, with the constraint that at least three modes must be included. The term ξ represents ~~the~~ residual error.

We impose statistical constraints, assuming that both the main effects β_i and interaction

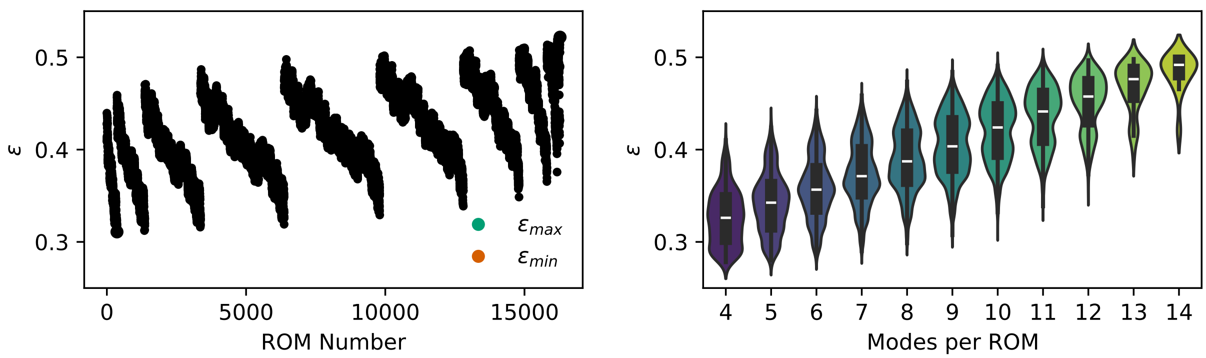

FIG. 19: Frequency and range-averaged coherence between detected wake centers from lidar scans and ROMs considering $f_s \leq St_{in}$ Hz (left). Distribution of ε by the number of modes contained in each ROM (right). ~~make vertical scales identical~~

terms γ_{ij} follow a normal distribution with zero mean, and that the residual error ξ follows a normal distribution. The goal is to estimate the coefficients β_i and γ_{ij} that minimize the residual error and provide insight into how modes and their combinations influence wake center coherence. This approach allows for a comprehensive sensitivity analysis, capturing the complex, potentially non-linear interactions between POD modes in representing wake center dynamics.

Figure 20 (right) shows the main effects β_i of each mode for the case beginning at 2023-09-04 10:30 UTC, the same case used to illustrate the POD modes above. One key finding illustrated by this figure is that the importance of each mode does not decay monotonically with the mode index. This indicates that although certain modes may represent more TKE in the full basis, they do not necessarily have a greater impact on wake meandering. The magnitude of the mode coefficients is small, $\beta \lesssim 0.1$, due to the bounded range of the average ROM coherence, $0.3 < \varepsilon < 0.55$. As shown in Eq. (15), β_i and γ_{ij} describe the linear and colinear dependence about the global mean coherence $\overline{\varepsilon}$, considering all 16,278 ROMs per case.

To illustrate the importance of each mode more completely, Fig. 21 splits the full distribution of ε (in white) into subsets of ROMs that include (blue) or exclude (orange) each of the first 14 modes. Modes with the strongest impact on ε (i.e., modes 1, 2, 3) show the greatest separation between distributions of ROMs that include or exclude ϕ_i. Moreover, modes that have the greatest positive impact on ε, contributing most to the accurate de-

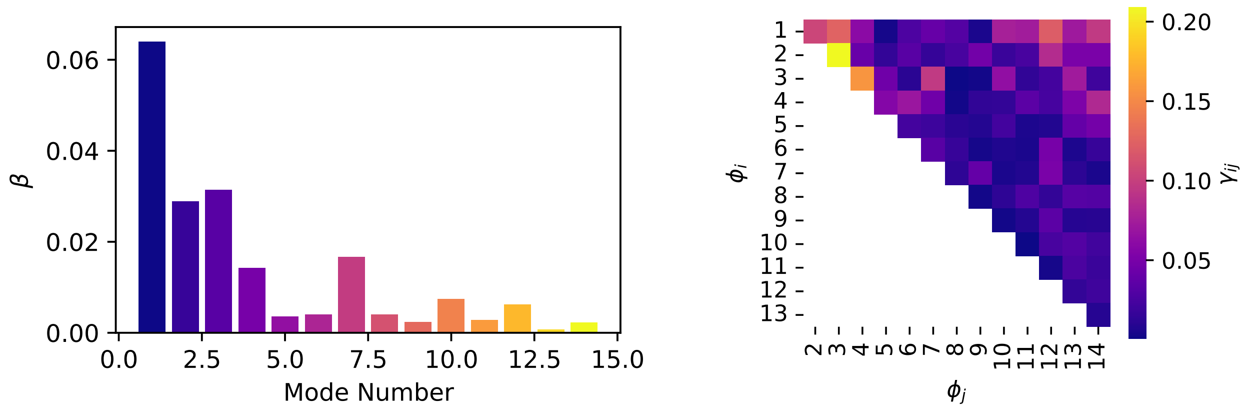

FIG. 20: Main effect β_i of each mode and interaction effects γ_{ij} between modes building the linear system that describes the dependence of average coherence on the presence or absence of each mode. **Does gamma only show absolute values??**

scription of wake meandering in the ROM, ~~$\hat{\mu}$,~~ always show distributions with means greater for ROMs that include ϕ_i compared to those that exclude ϕ_i. Conversely, distributions for modes with little influence on ε (e.g., modes 9, 11, 13) are nearly identical for ROMs that include or exclude the modes.

The case at 2023-09-04 10:30 UTC also exhibits relatively large mode interaction effects between a few particular mode pairs, as shown in Fig. 20 (right). In particular, ε shows a strong colinear dependence on modes 2 and 3, modes 3 and 4, and modes 1 and 3, in descending order. Figure 22 compares the distribution of ε for ROMs containing both modes in the pair $(\phi_i + \phi_j)$ against that for ROMs not containing both modes $\neg(\phi_i + \phi_j)$, including those that contain only one of the modes of interest. Interestingly, the distribution of ε for the pair $\phi_1 + \phi_3$ (right) is roughly normal, while for mode pairs $\phi_2 + \phi_3$ and $\phi_3 + \phi_4$ (left and center), the distributions are strongly bi-modal.

Verifying one of the key assumptions that underpins the ANOVA analysis described by Eq. (15), the residual ξ, shown in Fig. 23 (left), is roughly normally distributed. The range of ξ is relatively small, with a standard deviation of $\sigma_\xi \approx 0.003$, although the main effect β_i of some less relevant modes, falls within the range of the residual. Figure 23 (right~~0~~) also shows that the distributions of ξ, aggregated by the number of modes used to compose a ROM, are aapproximately normally distributed. Distributions of ANOVA residual appear to be consistently centered around $\xi = 0$, except for ROMs with 4 modes where there is a

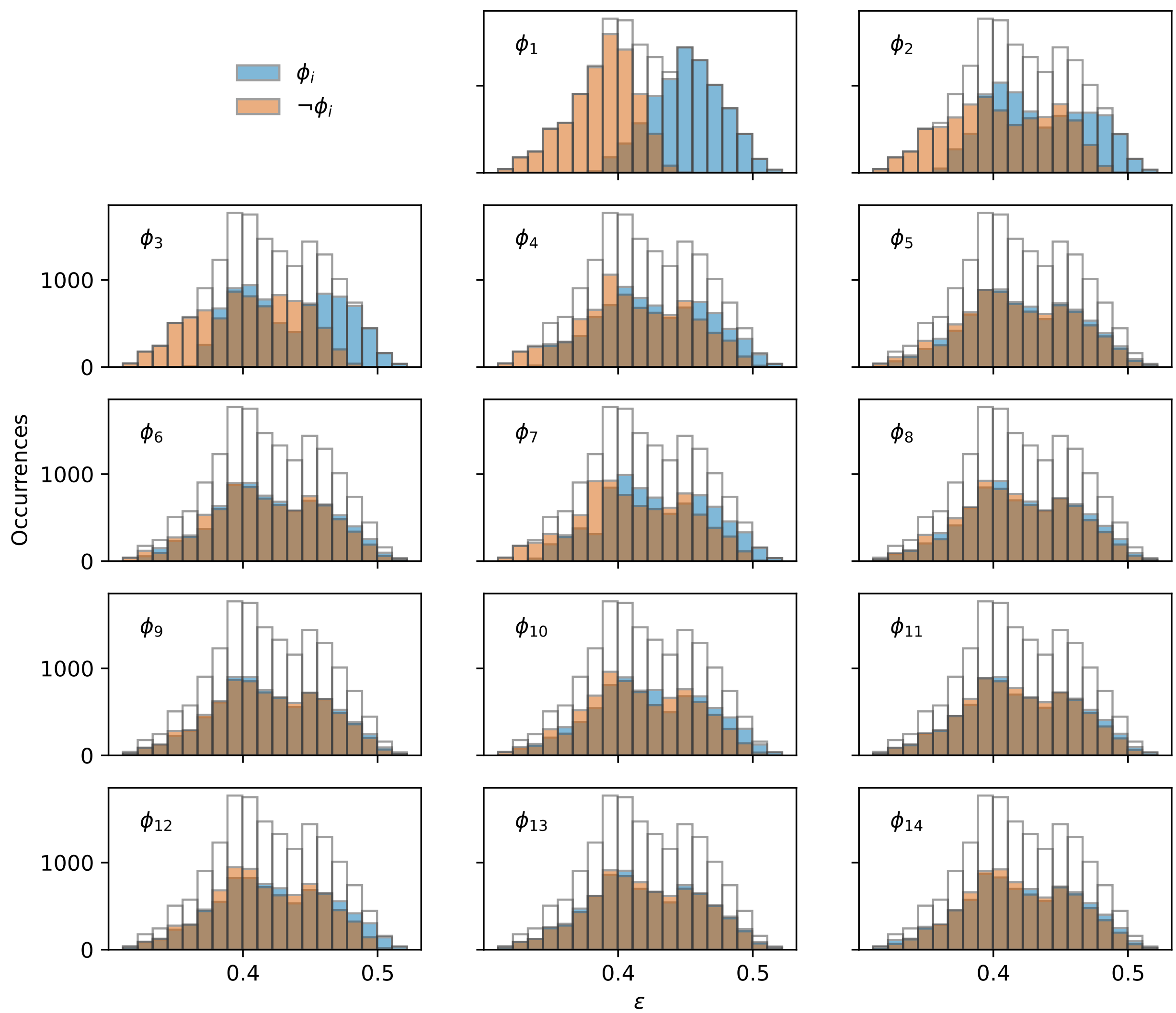

FIG. 21: Distributions of ε consider ROMs that include each mode (blue) or exclude each mode (orange).

slight negative bias, and ROMs with more than 11 modes for which there is a slight positive bias. Recalling that the sample size of ROMs composed with a given number of modes varies (Fig. 1), the distributions for ROMs composed with fewer than four or more than 11 modes are ~~not likely to be~~ statistaclly converged. Distributions of ROMs composed with bewteen 6 and 10 modes contain at least 1000 samples, suggesting that they represent distributions of ξ more accurately.

ADD figure that shows the wake reconstructions and detected wake centers for selected cases. Choose a few datetimes, compare lidar data to overall best and worst cases, the best ROM that require the fewest modes, and the worst

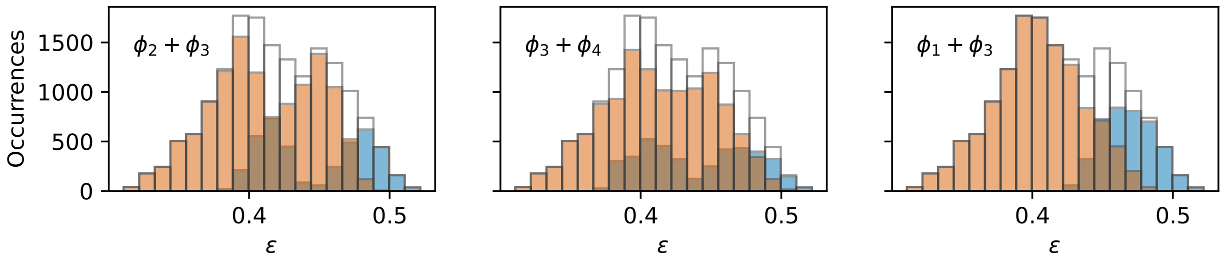

FIG. 22: Distributions of the average coherence for ROMs that include or exclude the indicated pairs of modes listed in descending order of importance. **For each mode pair, from left to right, $\gamma_{2,3} = XX$, $\gamma_{3,4} = XX$, and $\gamma_{1,3} = XX$.**

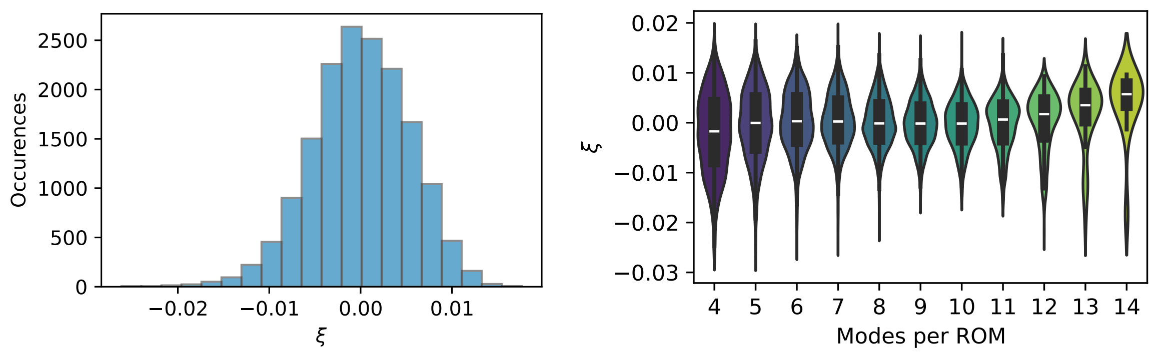

FIG. 23: Normally distributed of the ANOVA residual error, ξ (left), and residual error aggregated by the number of modes in each ROM (right).

ROM with the most modes. To illustrate the range of ROM accuracy, **??** compares wake measurements and detected wake centers from the lidar directlly to the low-order desciptions provided by selected modes. Specifically, **??** highlights the ROMs with the maximum and minimum values of ε, that is the ROMs that have wake center dynamics that show the maximum and minimum average correlation with the lidar observations.

Considering the full population of 65 cases ~~considered~~ in this study, Fig. 24 shows the overall trends of β with mode index. The mean profile in blue indicates that the first mode tends to contribute quite strongly to the linear model of ε, with a coefficeint of $\beta_1 = 0.4$. In most cases, the mode ϕ_1 is qualitatively similar to that shown for 2023-09-04 10:30 UTC in Fig. 15, exhibiting a strongly antisymettric character along the average wake centerline and a half-period of streawise variation. Figure 24 indicates that this mode can account for a huge amount of ε, ranging from a coeffcient of $0.2 \leq \beta_1 \leq 0.75$, as shown by the pink lines indicating maximum and minimum values.

The range of ϕ_2 is even larger, showing a maximum value of$\beta_2 = 0.72$, similar to that of β_1, while the minimum value is $\beta_2 = -0.02$, indicating that for some cases, the average correlations ε is weakly anticorrelated with mode ϕ_2. Similar behaviour is evident for modes ϕ_2 through ϕ_8, where at least the modes work against the succesful representation of ε for several cases.

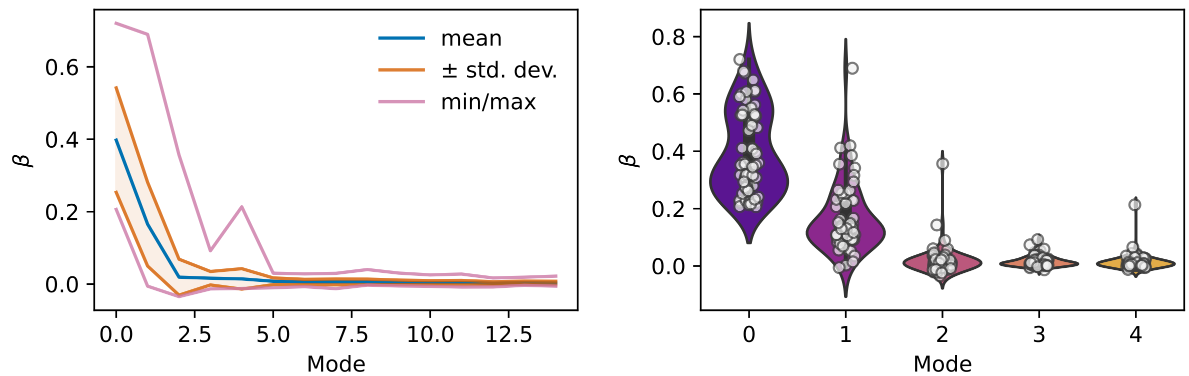

FIG. 24: Trends of β considering all 65 cases in the overall study. **Correct mode indices for both subplots. Should begin at 1...**

Figure 24 shows (right) the distributions of β_1 through β_4 is greater detail as violin plots with markers indicating values for individual cases. Overall, the figure suggests a trend of decreasing range of β as the mode increases. Modes 2, 3, and 5 show what appear to be outlier values extending the range to higher positive values of β. Mode 1 also appears to have abimodal distribion of β. Because there are only 65 cases in the current study, and those cases span a wide range of atmospheric conditions, insights draw from the relationship between ε and β_i should be treated with an appropriate level of caution.

The description of multi-colinear dependence of ε on the modal basis ϕ_i has a similar trend. Figure 25 shows the mean (left) and standard deviation (right) of γ_{ij} for the collection of 65 cases. Overall, ϕ_1 has the strongest colinear behvior with the other modes in the basis, with a maximum value of $\overline{\gamma}_{1,2} = 0.29$ and decreasing monotonically with j thereafter. The average colinear dependence of higher order modes is even lower, suggesting that the global behavior for ε may be that a purely linear realationshipo is sufficient for desciption of wake meandering. The standard deviation of $\sigma_{\gamma_{ij}}$ showns similar trends as the mean, where the largest values are located in the top left, for colinearity between ϕ_1 and ϕ_2, decreasing as

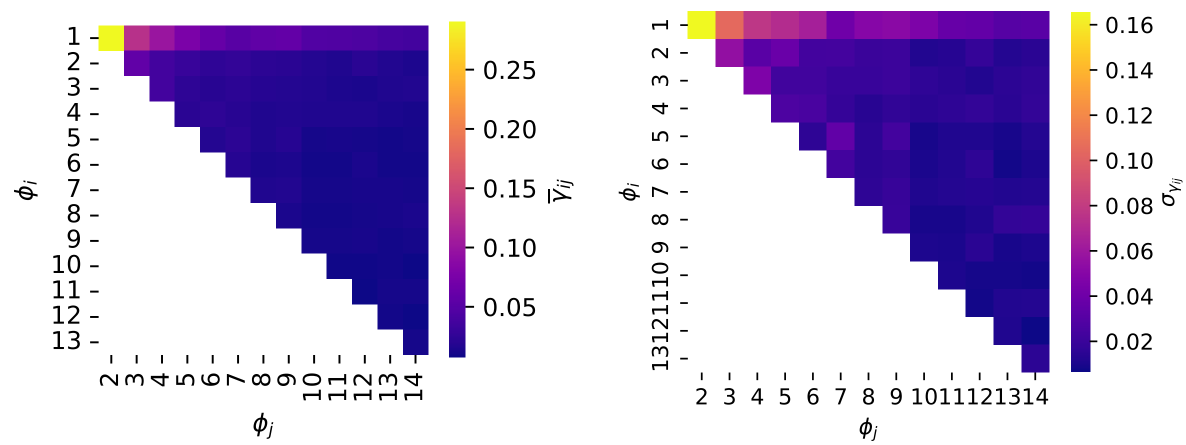

FIG. 25: Trends of $\overline{\gamma}_{ij} = f(\phi_i, \phi_j)$ considering all 65 cases in the overall study.**Does gamma only show absolute values??**

both i and j increase.

While the average behavior of β suggests a monotonic decrease in value mode index, the true relationship is likely more complex. Figure 26 compares the log of the mode eigenvalues,

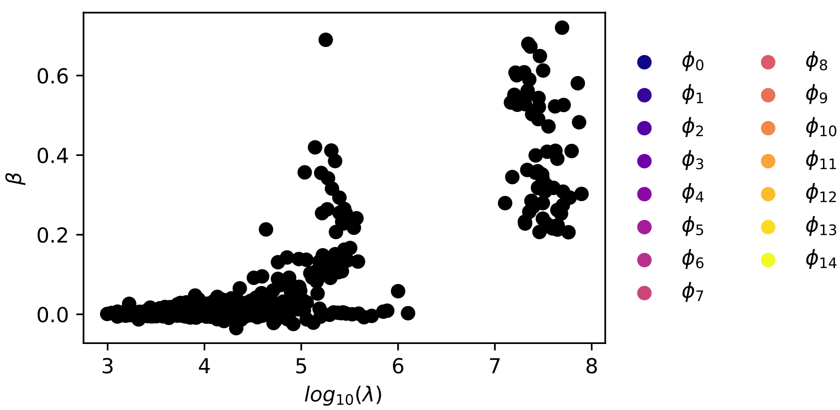

FIG. 26: Comparison of POD mode eigenvalues λ_i to main effect coefficients β_i for all 65 cases.

$\log_{10}(\lambda)$ to the main effect coefficients β, colored by the mode number. Because β_i tends toward zero for all cases, while λ_i show a wide range of values for each mode, no direct relationship between the two is directly evident in the current data. This disparity is thought to arise from the fact that the two variables descirbe related bute different phenomena. POD modes describe the variance-maximizing structures in the observational basis, that is the describe the TKE in the measurement domain of the nacelle-mounted scannig lidars—the eigenvalues λ_i quantify the TKE each mode represents in the overall budget. The main effect coefficients, β_i describe the dependence of the streamwise- and reduced frequency-averaged correlation between wake centers estimated by a Gaussian function fit to the observaitonal data and estimates made by a ROM as described by a subset of POD modes.

Make a figure that compares the dominant frequency of each mode to the inflow strouhal number for each case. Makers can be colored by the respective β coefficient. If there's sufficient evidence that modes with high β are correlated somehow, try to connect to MO length or other inflow condition.

V. CONCLUSIONS

This study leverages remote sensing data from nacelle-mounted scanning lidars to examine full-scale wind turbine wake dynamics under varying inflow conditions. By analyzing observational data from nacelle-mounted scanning lidars, we quantify wake meandering using snapshot POD and spectral methods. We find that the wake center‘s meandering statistics differ across test cases, revealing insights into the spatiotemporal and spectral characteristics of wake dynamics.

Our analysis of POD modes challenges conventional descriptions of wake meanding and the expected relationships with the atmospheric inflow. While low-order modes represent the largest, most energetic turbulent structures, they do not necessarily contribute most to accurate wake meandering representations. In fact, some modes show no correlation with meandering dynamics, while others distort or even detract from the accurate representation of wake meandering. This suggests that certain modes are irrelevant to wake meandering or are only relevant to specific regions.

Although spectral analysis of the mode coefficients correlated with meandering does not align with the expected Strouhal number relationships, the findings highlight important complexities. Specifically, wake meandering models based on either passive tracer dynamics or interactions between coherent structures fail to fully capture the observed dynamics. This discrepancy may stem from insufficient resolution of meandering structures in the measurements or a more intricate relationship between wake meandering and turbulence than current models suggest.

Althoguh quantitatively different from one case to another, the POD modes describing wake dynamics are qualitatively quite similar. This similarity suggests the existence of a semi-universal basis of functions or structures that describe wind turbine wakes over a wide range of atmospheric conditions. We suspect that a low-order manifold exists that efficiently descirbes wake aerodynamics and wake meandering in a way that can be comuptationally efficient to represent and could benefit steady-state and engineering wind turbine and wind plant wake models/

While this study is essentailly a phenomenological desciption of wake meandering and its relationship with turbulent structures describing the wake, it is not a dynamic or predictive model. One of the main limitations in our current approach is that changes in the azimuthal position of the wake from yaw activity of the turbine are not easily separated from the underlying wake aerodynamics. To bridge this gap, a similar workflow using the classical POD or dynamic mode decomposition could be applied to the timeseries of estiamted wake centers. By considering the additional dynamics of the turbulent atmospheric inflow and the operating state of turbine, a complete dynamical model for wake centerline trajectories, including wake meandering should be within reach.

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